PARISHRAM 2.0 2026 (Class 12th)

Mathematics

Relations and Functions

DPP: 02

Q1 Let A = {1, 2, 3} & B = {5, 7, 9} if a relation R is defined from A to B, defined as R = {(a, b) : a > b, $a \in A$ & b ∈ B} then R is
(A) Universal Relation
(B) Null Relation
(C) Identity Relation
(D) None of these

Q2 Let A = {–5, –2, –3} & B = {1 ,5, 7} if a relation R is defined from A to B defined by R = {(a, b) : a < b, a ∈ A & b ∈B} then R is
(A) Universal Relation
(B) Null Relation
(C) Identity Relation
(D) None of these

Q3 The void relation on a set A is
(A) Reflexive
(B) Symmetric and transitive
(C) Reflexive and symmetric
(D) Reflexive and transitive

Q4 The relation R in the set of natural number N defined as R = {(x, y) : y = x + 5 and x < 4} is:
(A) reflexive (B) transitive
(C) symmetric (D) None of these

Q5 If a relation R on the set {1, 2, 3} be defined by R = {(1, 2)}, then R is:
(A) reflexive (B) symmetric
(C) transitive (D) None of these

Q6 Let A = {1, 2, 3} and consider the relation R = {(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)}. Then R is:
(A) reflexive but not symmetric
(B) reflexive but not transitive
(C) symmetric and transitive
(D) neither symmetric nor transitive

Q7 What type of relation is R = {(1,3), (4,2), (2,4), (2,3), (3,1)} on the set A = {1, 2, 3, 4} ?
(A) Reflexive (B) Transitive
(C) Symmetric (D) None of these

Q8 Consider the non-empty set consisting of children in a family and a relation R defined as aRb, if a is brother of b. Then R is
(A) symmetric but not transitive
(B) transitive but not symmetric
(C) neither symmetric nor transitive
(D) both symmetric and transitive

Q9 If R is a relation on the set N, defined by $\{(x, y) : 2x - y = 10\}$, then R is
(A) reflexive
(B) symmetric
(C) transitive
(D) None of the above

Answer Key

Q1	B
Q2	A
Q3	B
Q4	B
Q5	C
Q6	A
Q7	D
Q8	B
Q9	D

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 **Text Solution:**

As no alement of set A is greater than the element of set B $\therefore$ R = ϕ

Video Solution:

Q2 **Text Solution:**

As every element of set A is less than every element of set B.

$\therefore (a, b) \in R$ for all $a \in A, b \in B$. Hence, R = A × B.

Video Solution:

Q3 **Text Solution:**

Void relation on a set A is symmetric, transitive but not reflextive

$\because$ there is no element present in the relation.

$\therefore (a, a) \notin R$ for all $a \in A$. Hence, not reflexive.

Also, since no element belongs to R, so, relation is symmetric and transitive trivially.

As there is no counter example which can prove that void relation is neither symmetric nor transitive.

Video Solution:

Q4 **Text Solution:**

R = {(1, 6), (2, 7), (3, 8)}

$\because (6, 6) \notin R$

$\Rightarrow$ R is not Reflexive.

For (2, 7) $\in$R

$\Rightarrow$ (2, 7) $\in$R but (7, 2)$\notin$R

$\Rightarrow$ R is not symmetric.

Clearly, R is transitive trivially as there is no counter example by which we can prove that relation R is not transitive.

Video Solution:

Q5 **Text Solution:**

R on the set {1, 2, 3} be defined by R = {(1, 2)}

It is clear that R is not reflexive as (1, 1)$\notin R$.

R is not symmetric as (1, 2)$\in R$ but (2, 1)$\notin R$.

Clearly, R is transitive trivially as there is no counter example by which we can prove that relation R is not transitive.

Video Solution:

Q6 **Text Solution:**

Given that, A = {1, 2, 3} and R = {(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)}

$\because$ (1, 1), (2, 2), (3, 3) $\in \mathbf{R}$

Hence, R is reflexive,

(1, 2) $\in$ R but (2, 1) $\notin$ R

Hence, R is not symmetric.

(a, b) $\in$ R and (b, c) $\in$ R

$\Rightarrow (\mathbf{a}, \mathbf{c}) \in \mathbf{R}$ for all a, b, c $\in \mathbf{R}$

Hence, R is transitive.

Video Solution:

Q7 **Text Solution:**

Reflexive:

Since, $(\mathbf{a}, \mathbf{a}) \notin \mathbf{R}$. Hence R on set A is not reflexive.

Symmetric:

Since $(2, 3) \in \mathbf{R}$ but $(3, 2) \in \mathbf{R}$. Hence R on set A is not symmetric.

Transitive:

Here,

$(1, 3) \in \mathbf{R}$ and $(3, 1) \in \mathbf{R}$ but $(1, 1) \notin \mathbf{R}$.

Hence, R on set A is not transitive relation.

Video Solution:

Q8 **Text Solution:**

Reflexive:

Here, $\mathbf{aRb} \Rightarrow \mathbf{a}$ is brother of b.

$\mathbf{aRa} \Rightarrow \mathbf{a}$ is a brother of a which is not true.

So, R is not reflexive.

Symmetric:

$\mathbf{aRb} \Rightarrow \mathbf{a}$ is a brother of b.

$\mathbf{bRa} \Rightarrow$ which is not true because b may be sister of a

Thus $\mathbf{aRb}$ does not imply $\mathbf{bRa}$. So, R is not symmetric.

Transitive:

Now, $\mathbf{aRb}, \mathbf{bRc} \Rightarrow \mathbf{aRc}$

a is the brother of b and b is the brother of c.

a is also the brother of c.

So, R is transitive.

Video Solution:

Q9 **Text Solution:**

Reflexive:

As $20 \in \mathbf{N}$ but $(20, 20) \notin \mathbf{R}$. So, R is not reflexive.

Symmetric:

As $(20, 30) \in \mathbf{R}$ but $(30, 20) \notin \mathbf{R}$. So, R is not symmetric.

Transitive:

As $(20, 30) \in \mathbf{R}, (30, 50) \in \mathbf{R}$ but $(20, 50) \notin \mathbf{R}$.

So, R is not transitive.

Video Solution: